**KEEP THESE INSTRUCTIONS FOR FUTURE REFERENCE**

# You printed a rocket. Now let's fly it!

This guide is for rockets made with **Rocket Builder** (rocket.quiverdesignworks.net). It shows what to do after the parts come off the printer: put them together, tie on the parachute, add a motor and launch.

Read each step all the way through before you start it. Work with a grown-up. Steps marked **ADULT STEP** are for the grown-up only.

The drawings show two sizes of rocket: the 18 mm (M18) Starter rocket, whose body is one print (the fin can D), and a taller 24 mm (M24) one whose body is three prints. Yours may have more or fewer body pieces. Drawings are not to scale.

## WHAT YOU NEED

NOT IN THE BOX: MOTOR, IGNITER, LAUNCH PAD AND CONTROLLER

- Fine sandpaper — 220–400 grit
- Hobby knife — adults only
- Medium super glue — CA, like Starbond Medium
- Ruler — in mm
- Pencil — to mark the balance point
- Scissors — cord, wadding
- Masking tape — fit fixes
- Stiff wire — to push the cord through
- Rubbing alcohol — cleans joints before gluing

## WHAT HAPPENS IN A FLIGHT

1. The igniter lights the motor. The rocket slides up the launch rod.
2. The motor burns out. The rocket coasts up to the top of its flight.
3. The motor's **ejection charge** puffs gas up the body. It pushes the wadding, the parachute and the nose out of the top.
4. The nose stays tied to the body by the **shock cord**. The parachute opens. Everything floats down together.

## HOW THE PARTS HOLD TOGETHER

- **Fins** are printed on the fin can. No fin gluing!
- **Body pieces** slide together on short shoulders. You glue these joints so they stay shut.
- **The nose** slides in too, but is **never glued**. It must pop off.
- **The motor** pushes against a ring at the top of its tube. The **motor cap** screws onto the tail and holds it in.

**SAFETY FIRST**

Model rocket motors and igniters are handled by an adult. Fly only in a big open field, with a launch pad and an electric launch controller, and follow the National Association of Rocketry (NAR) Model Rocket Safety Code (summary on page 8).

## STEPS AT A GLANCE

**1–3 Build** p. 3
Clean up, test-fit and glue the body.

**4–8 Recovery** p. 4–5
Tie on the cord and parachute, fold, pack.

**9–12 Motor and pad** p. 6–7
Adult steps: motor, cap, igniter, launch.

## MY ROCKET

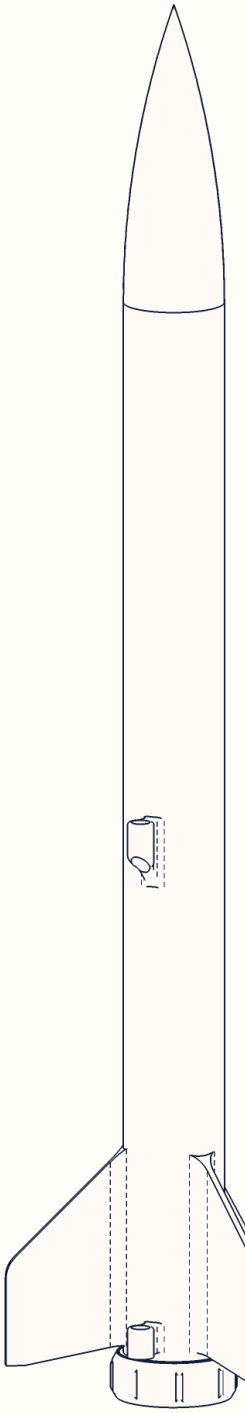
FILL THIS IN

NAME _____ BEST MOTOR _____

BUILDER _____ PARACHUTE _____

MOTOR SIZE _____ BALANCE POINT _____

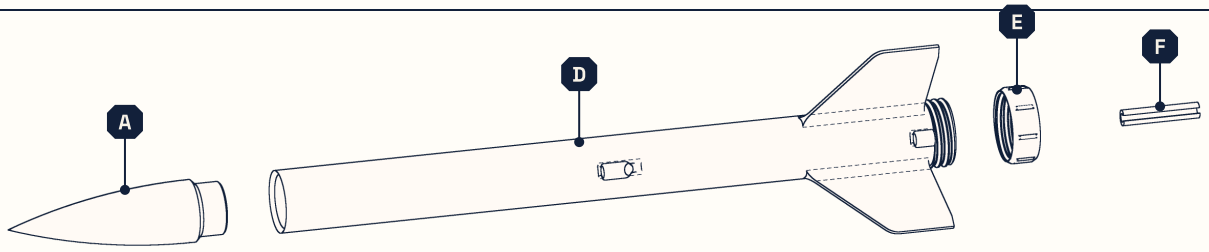
Copy these from your rocket's report in Rocket Builder or from *build_notes.txt* in your download. The balance point is the CG, in mm from the nose tip.



# Parts layout

Lay out your parts and check them off. Each part has a letter. The steps use the same letters.

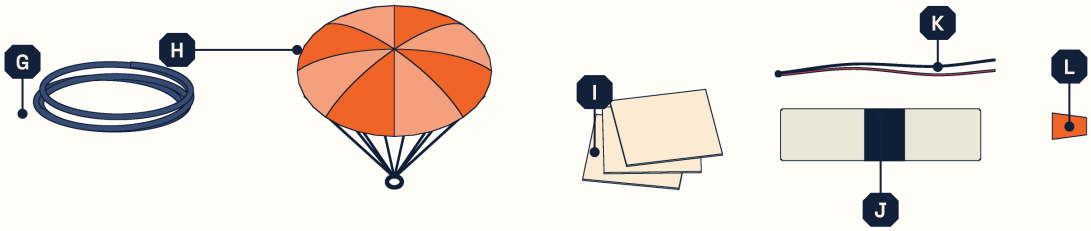
## PRINTED PARTS — A–F




A one-piece body, like the Starter and Mini kits. Taller rockets also have B and C.

| | PART | WHAT IT DOES | ✓ |
|---|---|---|---|
| A | **Nose cone** | Points the way. A short shoulder slides into the body. Inside the bottom is a little **tie bar** for the shock cord. | ☐ |
| B | **Top body section** (taller rockets only) | The top piece of a body printed in several pieces, with the cord **strap** (a bar across the inside) in its top. | ☐ |
| C | **Body section** (C1, C2… or none) | Middle pieces of a tall body, between B and D. A one-print body has no B or C. | ☐ |
| D | **Fin can** | Fins, launch lug(s), motor tube and the thread for the cap. On a one-print body (Starter, Mini) it's the whole tube, with the cord **strap** in its top. | ☐ |
| E | **Motor cap** | Screws onto the tail and holds the motor in. The hole lets the motor's flame out. | ☐ |
| F | **Rod standoff** | A clip for the launch rod. It stays on the pad and holds the rocket up off the blast plate. | ☐ |

## YOU ADD — G–L




| | | |
|---|---|---|
| G | **Shock cord** | Braided elastic, at least 3 times as long as the rocket, and never shorter than 36 in (90 cm). Flat 1/8 in (3 mm) wide for 13 and 18 mm motors, flat 1/4 in (6 mm) for 24 and 29 mm. Round 1/8 in elastic fits too. |
| H | **Parachute** (or streamer) | Use the size your Rocket Builder report suggests (“Which parachute?”). Light, narrow rockets use a streamer (the Mini kit has one). Kits include a **snap swivel** to clip it on with. |
| I | **Recovery wadding** | Flame-resistant tissue. It keeps the hot ejection gas off the parachute. Use new wadding every flight. |
| J | **Motor** ADULT | A certified model rocket motor of your rocket's size (M13, M18, M24 or M29) and length. Use the motor your report recommends. |
| K L | **Igniter and plug** ADULT | Come with the motors. The plug holds the igniter in the nozzle. |

**TIP**

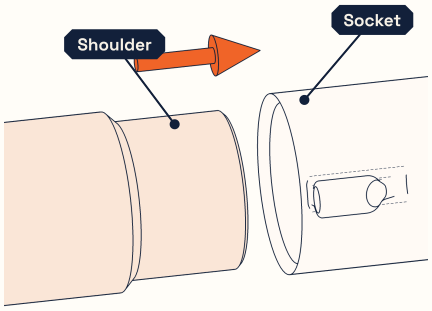
Not sure which motor or parachute? Your rocket's report in Rocket Builder shows the best motor, the parachute size and the launch field it needs.

# Build the rocket

## 1 Clean up the prints

- Pull off any brim, strings and blobs.
- The edge that sat on the print bed can bulge out a little. Sand that lip off the **shoulders** (the narrow ends) and the start of the **tail thread**.
- Look for cracks, gaps between layers or thin spots, especially where the fins meet the tube. Don't fly a cracked part. Print it again.

## 2 Test-fit the joints (no glue yet)

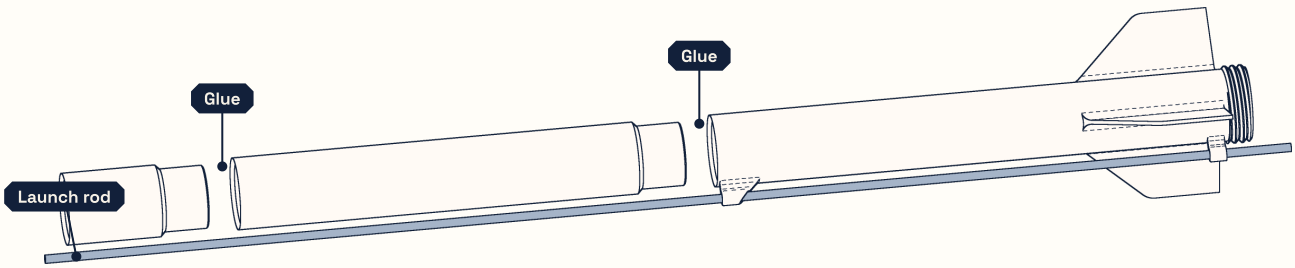



One-piece body (Starter, Mini): just push A into D. Taller rockets: push each shoulder into the next part with a little twist. It should go all the way in, snug, with no gap at the seam. Too tight? Sand the shoulder lightly and try again.

**NOSE FIT: THE NICKEL TEST (BEFORE ANY GLUING)**

Tie a scrap of cord to the nose's tie bar, push the nose into the body and hold it with the nose hanging down. Hang a small zip bag on the cord and add US nickels (5 g each) until the nose pulls out: **Mini (13 mm) about 2, Starter (18 mm) 4–5, Big (24 mm) 7–9.** Out sooner? Too loose: one wrap of tape on the nose's shoulder. Needs more? Too tight: sand the shoulder lightly. Other sizes: nose + bag + coins ≈ 74 g × the bore's area in square inches (29 mm: about 78 g).

## 3 Glue the body joints, with the launch lugs in line (taller rockets only)




One-piece bodies (Starter, Mini) skip this step: nothing to glue.

1. Sand the shoulder of C lightly, so it isn't shiny. Wipe it with rubbing alcohol. Let it dry.
2. **Grown-up:** put a thin ring of **medium super glue** round the shoulder, near its step.
3. Push it into D in **one smooth push**, all the way. Hold it 30 seconds. Don't stop halfway: the glue grabs in about 10 seconds.
4. Slide a launch rod through the lugs while the glue sets, so they line up. Glue B onto C (or onto D if you have no C) the same way.

**DO NOT GLUE**

the nose (A) or the motor cap (E). Keep glue out of the slot behind the cord strap. The fins are part of D: they need no glue.

**TIP**

Very tight joint? Push it together dry, then run a few drops of **thin** super glue round the seam: it soaks in. Printed in PETG? Super glue holds less well on PETG, so sand and clean it well.

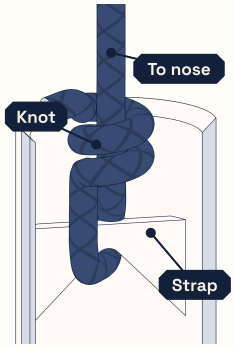
**SUPER GLUE SAFETY**

A grown-up opens the glue and does the gluing, with a window open and safety glasses on. It sticks skin in seconds. Stuck fingers? Don't pull. Soak them in warm soapy water and roll them apart gently. Glue near an eye: don't force it open; cover it with a wet pad and see a doctor. Activator spray is for grown-ups only, outdoors: it's flammable. Never spray it inside a joint before pushing it together.

# Rig the recovery system

The recovery system is the shock cord, the parachute and the wadding. It brings your rocket down gently, all in one piece.

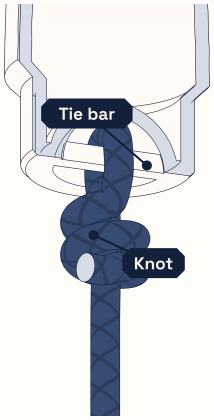
## 4 Tie the cord to the strap



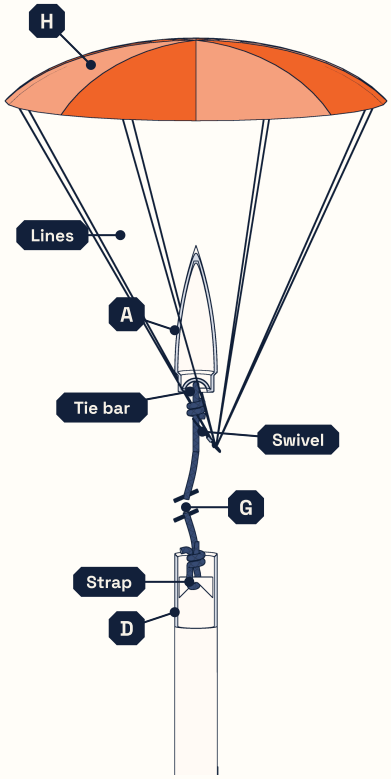

1. Take the nose off. Inside the top of the body (D, or B on taller rockets), the **strap** is a flat bar across one side, with a slot behind it.
2. Push one end of the elastic **down the slot behind the bar**. Pinch the end flat, or push it with a stiff wire. Pull it out under the bar.
3. Bring it up **in front of the bar**. Tie it round the long part with two overhand knots, close to the bar. Leave a tail as long as your thumb. No glue: it makes elastic brittle.

Or clip it on with a spare snap swivel through the slot behind the bar. The drawing has the front of the tube cut away.

## 5 Tie the other end to the nose

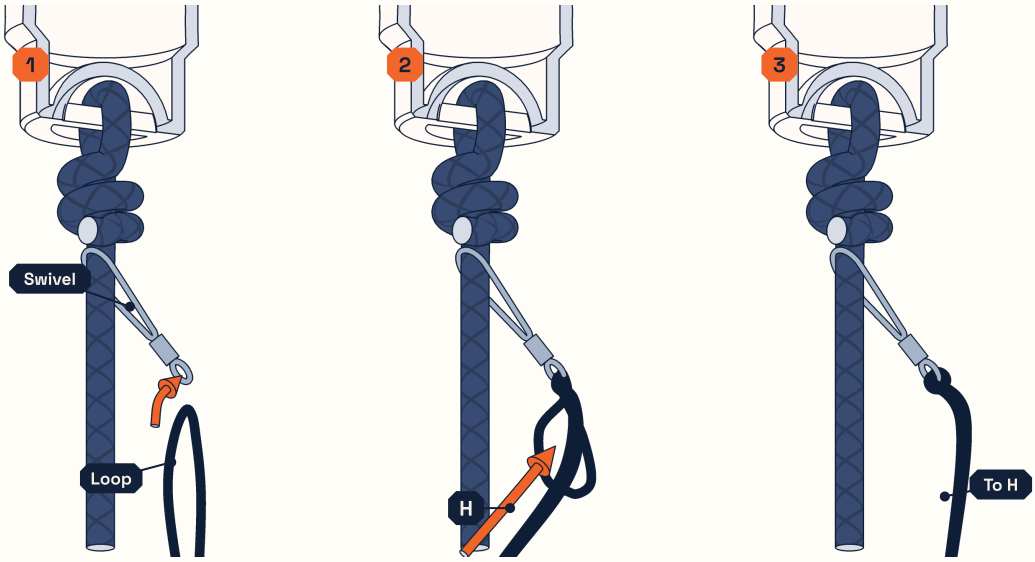



1. Turn the nose (A) over. A small **tie bar** crosses the dome in the bottom of its shoulder.
2. Push the cord's end up one side of the bar, over it and down the other side.
3. Tie it round the long part with two overhand knots, close to the nose. No glue.
4. Tug hard on the nose and the body. Nothing should slip.




How it hangs together in the air. **Not to scale:** the cord is really at least 3 times the rocket's length (the break mark). The nose and the top of D are drawn cut open.

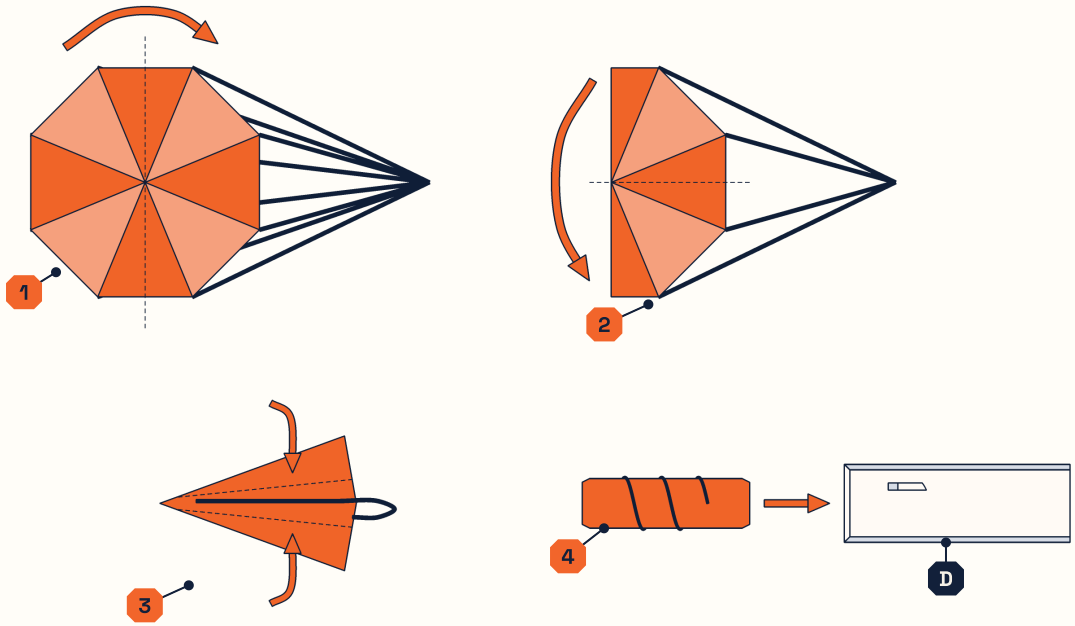
## 6 Clip the parachute on at the nose




1. Clip the **snap swivel** round the cord just below the nose knot. Hold the parachute's lines together by their ends so they make one **loop**.
2. Push the loop through the swivel's eye, then push the **whole parachute through the loop**.
3. **Pull snug** (a lark's head). To swap parachutes, just unclip the swivel and clip on another.

24 mm and bigger: clip the swivel to the nose's tie bar instead. 13 and 18 mm: the bar only has room for the cord, so clip it round the cord, as drawn (the knot stops it sliding). A streamer goes on the same way.

## 7 Fold and roll the parachute

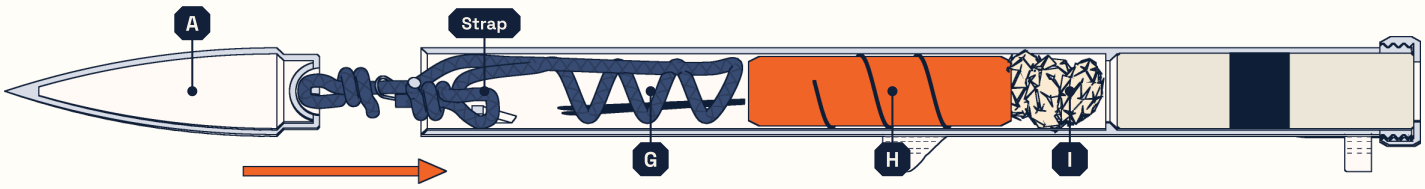



1. Lay it flat. Pull the lines straight and even. Fold it in half along the dashed line.
2. Fold it in half along the dashed line, then in half once more. Now it's a thin wedge and the lines meet in one bundle.
3. Lay the lines back along the middle of the wedge. Fold both sides over them along the dashed lines.
4. Roll it up loosely from the point. Wrap the lines once round it, loosely. It should slide in and out of the body easily (drawn next to the top of the body, real size).

**TIP**

Fold it fresh before every flight and store it unfolded. A pinch of talc keeps a plastic parachute from sticking on cold days.

## 8 Pack it: wadding, parachute, cord, nose




1. Loosely crumple wadding to fill about 1–2 body widths of the tube. Push it down until it sits on the ring at the top of the motor tube. Don't pack it hard.
2. Slide in the rolled parachute. Fold the cord in a zig-zag on top.
3. Push the nose on. Don't trap the cord or lines in the joint.
4. **Slide-out check** (first time): take the nose off and tip the rocket nose-down. The parachute and cord should slide out past the strap. If they stick, fold the parachute smaller or use a streamer.

**NOSE FIT**

You checked it with the nickel test (step 2). The nose should come off with a gentle pull. Never glue it on.

**NARROW ROCKETS: A STREAMER**

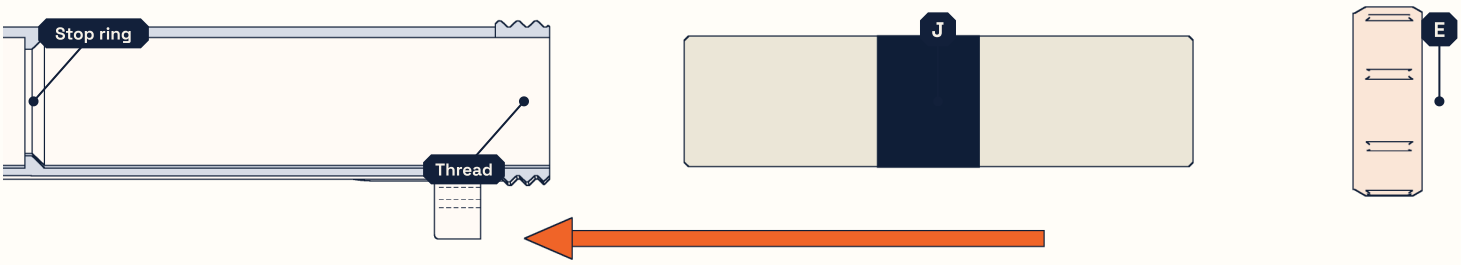
On a light rocket with a bore under 16 mm (like the Mini), use a **streamer** instead of a parachute: a parachute can jam past the strap. Rocket Builder suggests one, and the Mini kit comes with it.

# Motor, cap and igniter

**ADULT STEPS 9–12**

A grown-up handles the motor and igniter. Use only certified model rocket motors, as they come. Never cut, drill, heat or change a motor. Keep motors and igniters away from heat and sparks until launch time.

## 9 Slide the motor into the tail ADULT STEP

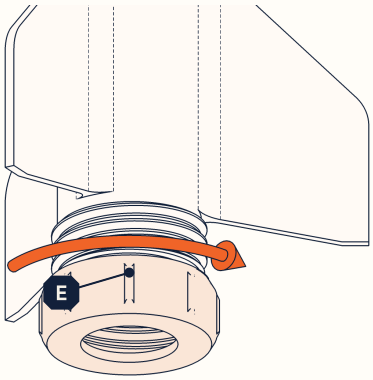



- Check the size: an 18 mm motor for an M18 rocket, and so on.
- Slide it in **nozzle end last**, so the nozzle points out of the tail.
- Push it all the way, until it stops against the ring at the top of the motor tube. That ring takes the motor's push.

**TOO TIGHT?**

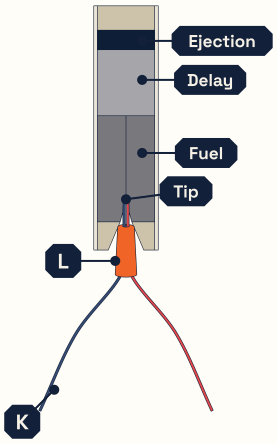
Never force a motor. Look for print strings inside the tube and sand them off. A full-length motor sticks out past the tail by about the height of the thread.

## 10 Screw on the motor cap ADULT STEP



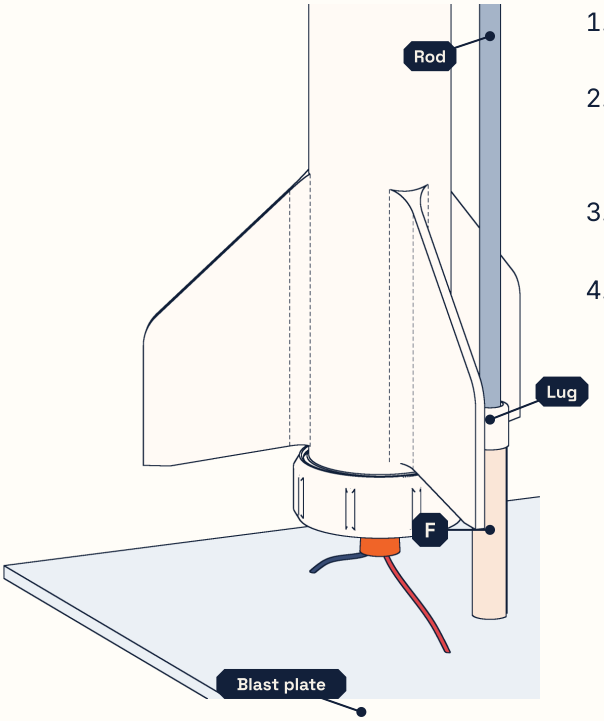

- Turn the cap onto the thread, clockwise as you look at the tail. Start it straight so it doesn't cross-thread.
- **Finger tight** is enough. The cap's floor rests on the end of the motor, so the motor can't be pushed out of the back when the ejection charge fires.
- Never glue or tape the cap on.

## 11 Put in the igniter ADULT STEP




1. Do this at the pad, with the safety key out of the controller.
2. Push the igniter (K) through the cap's hole into the nozzle, **all the way**, until it touches the fuel.
3. Hold it there and push in the plug (L).
4. Bend the two leads apart so they can't touch.

## 12 Set up on the pad




1. Snap the rod standoff (F) onto the rod, opening away from the rocket. Slide it down to the blast plate.
2. Slide the rocket down the rod by its lugs until the bottom lug rests on the standoff. The tail is now clear of the plate, with room for the clips.
3. Clip one controller clip to each igniter lead. Clips must not touch each other, the rod or the plate.
4. The rod points straight up (within 30°). Everyone steps back. Key in. Count down: **5, 4, 3, 2, 1, launch!**

### PREFLIGHT CHECKLIST — EVERY FLIGHT

- [ ] No cracks. Fins solid. Body joints glued all the way round.
- [ ] Cord tied tight at the strap and the nose. Tug both ends; replace a fuzzy, singed or stiff cord. Swivel clipped on, parachute hitched to it.
- [ ] New wadding, then the rolled parachute, then the cord. Nose snug, but it comes off.
- [ ] Right motor, pushed all the way in. Cap on, finger tight.
- [ ] **Balance test:** with the motor in, the rocket balances on your finger at the balance point from your report, or nearer the nose. Mark that point with pencil. If it balances behind the mark (nearer the tail), don't fly.
- [ ] The field is big enough (page 8). Wind under 20 mph. No low clouds or aircraft.
- [ ] Launch rod 1/8 in (3.2 mm) unless you changed it in Advanced. Blast plate on. Standoff on the rod.
- [ ] Dry grass and anything that burns cleared from around the pad.
- [ ] The adult in charge keeps the safety key. Everyone at least 15 ft back (30 ft for motors bigger than D).
- [ ] Everyone watching. Count down out loud.

**IF IT DOESN'T LAUNCH (MISFIRE)**
Take out the safety key. Wait 60 seconds before anyone walks to the rocket. Then check the igniter, or put in a new one.

# Fly safely

## SAFETY CODE, IN SHORT — NAR

- **Grown-ups handle motors and igniters.** Only certified, factory-made motors, used as the maker says.
- **Electric launch only:** a launch controller with a safety key and a switch that springs back to off.
- **Misfire?** Key out, then wait 60 seconds before anyone goes to the rocket.
- **Launch pad:** a rod or rail with a blast plate, within 30° of straight up. Cap the rod's tip when you're not launching.
- **Size limits:** at most 1500 g at liftoff and 320 N·s of total motor impulse.
- **Where:** outdoors, in an open field, away from buildings, power lines, roads and dry brush. Never launch at targets, into clouds, near aircraft, or with anything that burns as a payload.
- **Recovery:** always a parachute or streamer, with flame-resistant wadding. Never climb trees or power poles to get a rocket back.

This is a summary. Read the full code at nar.org/ModelRocketSafetyCode.

## LAUNCH FIELD SIZE — NAR MINIMUM, EACH SIDE

| MOTOR CLASS | FEET | METRES |
|---|---|---|
| 1/4A, 1/2A | 50 | 15 |
| A | 100 | 30 |
| B | 200 | 60 |
| C | 400 | 120 |
| D | 500 | 150 |
| E, F | 1000 | 300 |

On a breezy day a rocket can drift further than this. Your Rocket Builder report gives the size your rocket needs (“How big a launch field?”). Put the pad near the upwind edge.

## AFTER THE FLIGHT

- Let the motor cool. Unscrew the cap. Push the used motor out from the front with a dowel.
- Shake out the burnt wadding. Wipe soot off the cap and thread.
- Check the cord, knots, parachute and joints. Reprint a cracked cap or part.

## TROUBLESHOOTING

| PROBLEM | TRY THIS |
|---|---|
| A shoulder or the nose is too tight | Sand off the bulge at its bottom edge (it printed on the bed), then sand lightly all round. Test often. |
| The nose falls off too easily | One wrap of masking tape round the nose's shoulder. Add more only if needed. |
| The cord won't go behind the strap | Pinch the end flat, or push it with a stiff wire. Clear print strings out of the slot. The slot fits 1/8 in round elastic or flat elastic folded double; thicker cord won't go. |
| The motor won't go in | Is it the right size? Clear strings from the motor tube. Never force it. |
| The cap won't screw on | Is the motor all the way in? Clean the first turn of the thread. Start the cap straight. |
| The motor didn't light | Key out, wait 60 s. Check the igniter tip touches the fuel and the clips are clean. Try a new igniter. |
| The parachute didn't open, or melted | More wadding. Roll the parachute looser. Check the motor's delay (the last number) matches your report. |
| The rocket wobbled or looped | Do the balance test. Check the score and motor in Rocket Builder. Don't fly again until it's fixed. |
| The body came apart at a joint | That joint wasn't glued all the way round. Clean it and glue it again. |

## FLIGHT LOG

| DATE | MOTOR | WEATHER | HOW IT FLEW / NOTES |
|---|---|---|---|
| | | | |
| | | | |
| | | | |
| | | | |